

Model Question Paper-I (CBCS Scheme)

USN

Third Semester B.E Degree Examination

AV-MATHEMATICS FOR EC ENGINEERING STREAM (BMATEC301)

TIME: 03 Hours

Max.Marks:100

Note: (i) Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.
(ii) Statistical tables and Mathematics Formula handbooks are allowed.

Module -1			M	L	C																
Q.01	a	Find the Fourier series expansion of the function $f(x)= x $ in $(-\pi, \pi)$.	6	L2	CO1																
	b	Find the half range Fourier sine series for $f(x)=\begin{cases} \frac{1}{4}-x & \text{in } 0 < x < \frac{1}{2} \\ x-\frac{3}{4} & \text{in } \frac{1}{2} < x < 1 \end{cases}$	7	L2	CO1																
	c	Find the constant term and the first harmonics in the Fourier series of f(x) given <table border="1"><tr><td>x</td><td>0</td><td>$\frac{\pi}{3}$</td><td>$\frac{2\pi}{3}$</td><td>π</td><td>$\frac{4\pi}{3}$</td><td>$\frac{5\pi}{3}$</td><td>2π</td></tr><tr><td>f(x)</td><td>1.0</td><td>1.4</td><td>1.9</td><td>1.7</td><td>1.5</td><td>1.2</td><td>1.0</td></tr></table>	x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π	f(x)	1.0	1.4	1.9	1.7	1.5	1.2	1.0	7	L3	CO1
x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π														
f(x)	1.0	1.4	1.9	1.7	1.5	1.2	1.0														
OR																					
Q.02	a	Obtain the Fourier series of the square wave given by $f(x)=\begin{cases} -k & \text{in } -\pi < x < 0 \\ k & \text{in } 0 < x < \pi \end{cases}$	6	L2	CO1																
	b	Find the half range cosine series for $f(x)=\pi-x$ in $(0, \pi)$. Hence deduce that $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+...=\frac{\pi^2}{8}$	7	L2	CO1																
	c	The following table gives the variations of a periodic current A over a period T: <table border="1"><tr><td>t (sec)</td><td>0</td><td>T/6</td><td>T/3</td><td>T/2</td><td>2T/3</td><td>5T/6</td><td>T</td></tr><tr><td>A (amp)</td><td>1.98</td><td>1.30</td><td>1.05</td><td>1.30</td><td>-0.88</td><td>-0.25</td><td>1.98</td></tr></table> Show that there is a constant part of 0.75 amps in the current A, and obtain the amplitude of the first harmonic.	t (sec)	0	T/6	T/3	T/2	2T/3	5T/6	T	A (amp)	1.98	1.30	1.05	1.30	-0.88	-0.25	1.98	7	L3	CO1
t (sec)	0	T/6	T/3	T/2	2T/3	5T/6	T														
A (amp)	1.98	1.30	1.05	1.30	-0.88	-0.25	1.98														
Module-2																					
Q. 03	a	Find the Fourier transform of $f(x)=\begin{cases} 1- x , & x \leq a \\ 0, & x >a \end{cases}$. Hence evaluate $\int_0^\infty \frac{\sin^2 t}{t^2} dt$	6	L3	CO2																

	b	Find the Fourier cosine transform of $f(x) = \begin{cases} x & ; & 0 < x < 1 \\ 2 - x & ; & 1 < x < 2 \\ 0 & ; & x > 2 \end{cases}$	7	L2	CO2
	c	i) Find the Discrete Fourier Transform of the signal $f = [3, 4, 5, 5]^T$. ii) Find the Inverse Discrete Fourier Transform of the signal obtained in part (i)	7	L3	CO2
OR					
Q.04	a	Find the Fourier transform of $f(x) = \begin{cases} a^2 - x^2, & x \leq a \\ 0, & x > a \end{cases}$ where a is a positive constant	6	L2	CO2
	b	Find the Fourier sine and cosine transform of $f(x) = e^{-\alpha x}, \alpha > 0$.	7	L2	CO2
	c	Solve the integral equation $\int_0^\infty f(\theta) \cos \alpha \theta d\theta = \begin{cases} 1 - \alpha, & 0 \leq \alpha \leq 1 \\ 0, & \alpha > 1 \end{cases}$ Hence evaluate $\int_0^\infty \frac{\sin^2 t}{t^2} dt$.	7	L3	CO2
Module-3					
Q. 05	a	Find the Z- Transforms of a) $(n + 1)^2$ b) $\sin n\theta$	6	L2	CO3
	b	Find the inverse z-transforms of $\frac{8z - z^3}{(4 - z)^3}$	7	L3	CO3
	c	Solve the difference equation of $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$ with $y_0 = y_1 = 0$ using Z-transforms.	7	L3	CO3
OR					
Q. 06	a	Find the Z-transforms of a) $\sin(3n + 5)$ b) $\cos\left(\frac{n\pi}{2} + \frac{\pi}{4}\right)$	6	L1	CO3
	b	Obtain the inverse z-transform of $\frac{2z^2 + 3z}{(z+2)(z-4)}$	7	L3	CO3
	c	If $U(z) = \frac{2z^2 + 5z + 14}{(z-1)^4}$, evaluate u_2 and u_3 .	7	L3	CO3
Module-4					
Q. 07	a	Solve $\frac{d^3 y}{dx^3} + y = \cos 2x$	6	L2	CO4
	b	Solve $(D^2 - 4D + 3)y = (e^x + 1)^2$	7	L2	CO4
	c	Solve $x^2 \frac{d^2 y}{dx^2} - 4x \frac{dy}{dx} + 6y = \sin(2 \log x)$	7	L3	CO4
OR					

Q. 08	a	Solve $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 25y = e^{2x} + \sin x + x$	6	L2	CO4																																	
	b	Solve $(2x-1)^2 \frac{d^2y}{dx^2} + (2x-1)\frac{dy}{dx} - 2y = 8x^2 - 2x + 3$.	7	L2	CO4																																	
	c	In an LCR circuit the charge q on a plate of a condenser is given by $L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{c} = E \sin pt$. The circuit is tuned to resonance so that $p^2 = \frac{1}{LC}$. If initially the current I and the charge q are zero, show that for small values of $\frac{R}{L}$, the current in the circuit at time t is given by $\frac{Et}{2L} \sin pt$.	7	L3	CO4																																	
Module-5																																						
Q. 09	a	An experiment gave the following values: <table><tr><td>v(ft/min)</td><td>350</td><td>400</td><td>500</td><td>600</td></tr><tr><td>t(min.)</td><td>61</td><td>26</td><td>7</td><td>26</td></tr></table> It is known that v and t are connected by the relation $v = ab^t$. Find the best possible values of a and b.	v(ft/min)	350	400	500	600	t(min.)	61	26	7	26	6	L2	CO5																							
v(ft/min)	350	400	500	600																																		
t(min.)	61	26	7	26																																		
	b	In a partially destroyed laboratory record, the lines of regression of y on x and x on y are available as $4x - 5y + 33 = 0$ and $20x - 9y = 107$. Calculate $\bar{x}$, $\bar{y}$ and the coefficient of correlation between x and y.	7	L3	CO5																																	
	c	Ten competitors in a beauty contest are ranked by two judges A and B in the following order: <table><tr><td>ID No. of competitors</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>Judge A</td><td>1</td><td>6</td><td>5</td><td>10</td><td>3</td><td>2</td><td>4</td><td>9</td><td>7</td><td>8</td></tr><tr><td>Judge B</td><td>6</td><td>4</td><td>9</td><td>8</td><td>1</td><td>2</td><td>3</td><td>10</td><td>5</td><td>7</td></tr></table> Calculate the rank correlation coefficient.	ID No. of competitors	1	2	3	4	5	6	7	8	9	10	Judge A	1	6	5	10	3	2	4	9	7	8	Judge B	6	4	9	8	1	2	3	10	5	7	7	L3	CO5
ID No. of competitors	1	2	3	4	5	6	7	8	9	10																												
Judge A	1	6	5	10	3	2	4	9	7	8																												
Judge B	6	4	9	8	1	2	3	10	5	7																												
OR																																						
Q. 10	a	Fit a parabola $y = ax^2 + bx + c$ for the data <table><tr><td>x</td><td>1.0</td><td>1.5</td><td>2.0</td><td>2.5</td><td>3.0</td><td>3.5</td><td>4.0</td></tr><tr><td>y</td><td>1.1</td><td>1.3</td><td>1.6</td><td>2.0</td><td>2.7</td><td>3.4</td><td>4.1</td></tr></table>	x	1.0	1.5	2.0	2.5	3.0	3.5	4.0	y	1.1	1.3	1.6	2.0	2.7	3.4	4.1	6	L2	CO5																	
x	1.0	1.5	2.0	2.5	3.0	3.5	4.0																															
y	1.1	1.3	1.6	2.0	2.7	3.4	4.1																															
	b	The following table gives the heights of fathers(x) and sons (y): <table><tr><td>x</td><td>65</td><td>66</td><td>67</td><td>67</td><td>68</td><td>69</td><td>70</td><td>72</td></tr><tr><td>y</td><td>67</td><td>68</td><td>65</td><td>68</td><td>72</td><td>72</td><td>69</td><td>71</td></tr></table> Find the lines of regression and hence calculate the coefficient of correlation.	x	65	66	67	67	68	69	70	72	y	67	68	65	68	72	72	69	71	7	L3	CO5															
x	65	66	67	67	68	69	70	72																														
y	67	68	65	68	72	72	69	71																														

	c	Find the rank correlation for the following data:												7	L3	CO5
		x:	56	42	72	36	63	47	55	49	38	42	68			
		y:	147	125	160	118	149	128	150	145	115	140	152			